Capstone Lab 1

This capstone lab covers topics learned in Chapters 1 through 4 of *Guide to TCP/IP: IPv6 and IPv4*. It assumes you are working in a Windows 7 Professional or Windows 10 Pro environment. See the Preface for details about system requirements. To complete this lab, you must have Wireshark for Windows software installed and the trace (data) files named capstone-labs-1-trace-file1.pcapng and capstone-labs-1-trace-file2.pcapng.

Capstone Lab 1.1: Create Wireshark Display Filters to View Only DHCP Discover Messages

Time Required: 15 minutes

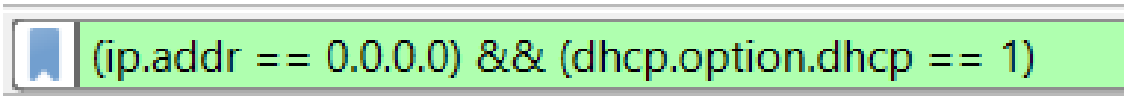
Objective: Use Wireshark to view a trace file and configure display filters to view only DHCP Discover messages from a specific host.

Description: In this project, you will create display filters in Wireshark and apply them to a trace file to view DHCP messages. The use of display filters allows for easier and often quicker viewing of a specific component in large trace files and/or real-time capturing of data. You need the capstone-labs-1-trace-file1.pcapng file to complete the lab.

1. Start Wireshark and open the **capstone-labs-1-trace-file1.pcapng** file.
2. Configure a display filter to find all packets that have an IPv4 source or destination address of 0.0.0.0 and apply the filter to the trace file.
3. Browse the list of IPv4 packets with an address of 0.0.0.0 and select a packet that has "DHCP Discover" in the Info column.
4. In the packet details pane, expand **Bootstrap Protocol (Discover)**.
5. Expand **Option: (53) DHCP Message Type (Discover)**, right-click **DHCP: Discover (1)**, select **Apply as Filter**, and then select **…and Selected**.
6. What did you accomplish in this part of Capstone Lab 1? Summarize your answer.
7. Clear the display filter box.

Submitted by Hershey Beatrice Marbeda

Question 5 – Please provide a screenshot



Question 6 – Please answer the lab summary question.

I learned how to find an IP address with 0.0.0.0 and choose the Discovery option. I also used Apply as Filter with 'and Selected'.

Capstone Lab 1.2: Create Wireshark Display Filters to View Only DHCPv6 Solicit Messages

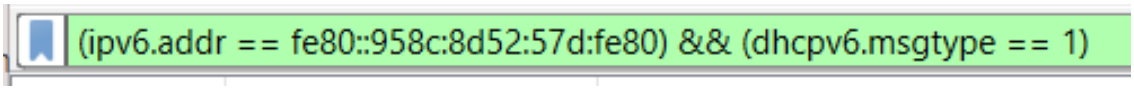
Time Required: 15 minutes

Objective: Use Wireshark to view a trace file and configure display filters to view only DHCPv6 Solicit messages from a specific host.

Description: In this project, you will use Wireshark to configure specific display filters to view individual DHCPv6 messages. You need the capstone-labs-1-trace-file1.pcapng file to complete the lab.

1. In Wireshark, open the **capstone-labs-1-trace-file1.pcapng** file, if it's not already open.
2. Configure a display filter to find all packets that have an IPv6 address of fe80::958c:8d52:57d:fe80.
3. Once you have found all the IPv6 packets with a source or destination address of fe80::958c:8d52:57d:fe80, look through the list and select a packet that has "DHCPv6" in the Protocol column and "Solicit" in the Info column.
4. In the packet details pane, expand **DHCPv6**.
5. Right-click **Message type: Solicit (1)**, select **Apply as Filter**, and then select **...and Selected**.
6. What did you accomplish in this part of Capstone Lab 1? Summarize your answer.
7. Clear the display filter box.

Question 5 – Please provide screenshot



Submitted by Hershey Beatrice Marbeda

Question 6 – Please answer lab summary question.

In this lab, I learned how to use Wireshark filter options to display specific IPv6 traffic using the address fe80::958c:8d52:57d and to filter DHCPv6 packets by Message Type 1, which represents a Solicit message.

Capstone Lab 1.3: Create a Wireshark Display Filter to View OSPF Packets

Time Required: 15 minutes

Objective: Use Wireshark to view a trace file and configure display filters to view OSPF (versions 2 and 3) packets.

Description: In this project, you will use Wireshark to configure display filters to view all OSPF packets, which include both OSPFv2 and OSPFv3. You need the capstone-labs-1-trace-file1.pcapng file to complete the lab.

1. In Wireshark, open the **capstone-labs-1-trace-file1.pcapng** file, if it's not already open.
2. Configure a display filter to find all OSPF packets and then select one of the OSPF packets.
3. Expand **Open Shortest Path First** in the packet details pane.
4. Where can you find the version of OSPF that's referenced in this packet?
5. Look through the OSPF packets listed. Viewing those from IPv4 and IPv6 addresses, determine which OSPF version relates to an IPv4-sourced OSPF packet and an IPv6-sourced OSPF packet. Write this information down.
6. Close Wireshark, and do not select **Save** if prompted to save changes upon closing.
7. What did you accomplish in this part of Capstone Lab 1? Summarize your answer.

Question 5 – Please provide screenshot showing this information, rather than writing it down.



IPv6 is version 3 and IPv4 is 2

Question 7 – Please answer lab summary question.

Submitted by Hershey Beatrice Marbeda

In this lab, I learned how to filter and display all OSPF packets in Wireshark. I also learned how to identify the OSPF header and distinguish between OSPFv2, which is used with IPv4, and OSPFv3, which is used with IPv6.

Capstone Lab 1.4: View IPv4 and IPv6 Headers for Specific Information

Time Required: 15 to 30 minutes

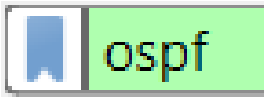
Objective: Use Wireshark to view a trace file and compare Differentiated Services Code Point (DSCP) values in OSPFv2 for IPv4 and OSPFv3 for IPv6 packets.

Description: In this project, you will use Wireshark to view header information in a supplied trace file. Specifically, you will view and compare DSCP values in OSPFv2 for IPv4 and OSPFv3 for IPv6 packets. You need the capstone-labs-1-trace-file2.pcapng file to complete the lab.

1. Start Wireshark and open the **capstone-labs-1-trace-file2.pcapng** file.
2. Configure a display filter for ospf.
3. Look through the list of packets and select one that has an IPv4 address in the Source column and has "Hello Packet" in the Info column.
4. Expand **Internet Protocol Version 4** in the packet details pane.
5. Expand **Differentiated Services Field**. View and make note of the Differentiated Services Codepoint value.
6. In the packet list pane, select a packet that has an IPv6 address in the Source column and has "Hello Packet" in the Info column.
7. Expand **Internet Protocol Version 6** in the packet details pane.
8. Expand **Traffic class**. View and make note of the Differentiated Services Codepoint value.
9. Repeat steps 3-8, but this time pay attention to which IPv4 and IPv6 source addresses are from the same router or switch. (For example, source addresses of 10.0.0.250 and fe80::f250 are from the same router.) This trace file contains traffic associated with four routers/switches from three different manufacturers. One brand of router/switch has different DSCP values between OSPFv2 and OSPFv3 Hello Packets. What is the equipment? Identify the equipment by IP address and manufacturer.
10. What did you accomplish in this part of Capstone Lab 1? Summarize your answer.
11. Close Wireshark, and do not select **Save** if prompted to save changes upon closing.

Submitted by Hershey Beatrice Marbeda

Question 8 – Please provide screenshots showing the different router and switches.



Differentiated Services Codepoint: Class Selector 6 (48)

Question 9 – Please answer question.

IP Address	Manufacturer	MAC	DiffServ Code
10.0.0.250	Cisco	00:1f:6c:cf:6e:50	48
fe80::f250	Cisco	00:1f:6c:cf:6e:50	56
10.0.0.251	Cisco	00:11:21:54:4b:c0	48
fe80::f251	Cisco	00:11:21:54:4b:c0	56
10.0.0.252	ProCurve	00:24:a8:db:ef:c0	48
fe80::f252	ProCurve	00:24:a8:db:ef:c0	48
10.0.0.254	3Com	40:01:c6:67:bc:01	48
fe80::f254	3Com	40:01:c6:67:bc:01	48

Cisco had 48 for V2 and 56 for V3; for 3 addresses would end with 250 and 251

Question 10 – Please answer lab summary question.

Submitted by Hershey Beatrice Marbeda

In this lab, I learned how to identify and compare DiffServ Code values in OSPF packets. I observed that the Cisco devices used a DiffServ Code of 48 for OSPFv2 over IPv4 and 56 for OSPFv3 over IPv6, while the ProCurve and 3Com devices used a value of 48 for both IPv4 and IPv6 traffic. This helped me understand how packet header values can differ depending on the protocol version and device.

Capstone Lab 1.5: View IPv6 Next Headers

Time Required: 15 minutes

Objective: Use Wireshark to view a trace file and compare Next header values in IPv6 packets.

Description: In this project, you will use Wireshark to view header information using a supplied trace file. Specifically, you will view and compare Next header values in IPv6 packets. You need the capstone-labs-1-trace-file1.pcapng file to complete the lab.

1. Start Wireshark and open the **capstone-labs-1-trace-file1.pcapng** file.
2. Configure a display filter for ipv6.nxt.
3. Select a packet.
4. Expand **Internet Protocol Version 6** in the packet details pane.
5. View and make note of the Next header value and the Layer 4 protocol associated with it.
6. In the packet list pane, select a different packet.
7. View and make note of the Next header value and the Layer 4 protocol associated with it.
8. Continue to repeat steps 6 through 7 for additional protocol type traffic. You should have a list of approximately 11 different protocols; however, some protocols share the same Next header value.
9. What did you accomplish in this part of Capstone Lab 1? Summarize your answer.
10. Close Wireshark, and do not select **Save** if prompted to save changes upon closing.

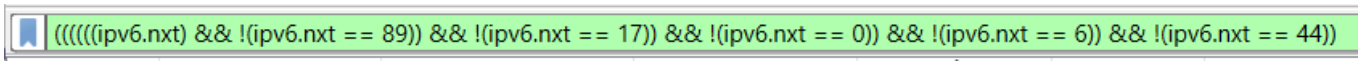
Question 5 – Please provide screenshot showing the next header and layer 4 protocol.

- Next Header: ICMPv6 (58)
- Next Header: OSPF IGP (89)
- Next Header: UDP (17) DNS TCP LLMNR DHCPv6 UDP/XML
- Next Header: IPv6 Hop-by-Hop Option (0) ICMPv6
- Next Header: TCP (6) TCP TELNET SSHv2 HTTP

Submitted by Hershey Beatrice Marbeda

- Next Header: Fragment Header for IPv6 (44)
- Next Header: Fragment Header for IPv6 (44) IPv6 ICMPv6

Question 7 – Please provide screenshot showing the next header and layer 4 protocol for a different packet.



((((((ipv6.nxt) && !(ipv6.nxt == 89)) && !(ipv6.nxt == 17)) && !(ipv6.nxt == 0)) && !(ipv6.nxt == 6)) && !(ipv6.nxt == 44))

Question 9 – Please answer lab summary question.

In this lab, I learned how to identify the IPv6 Next Header field and determine which protocol comes next in a packet. I analyzed different packets containing protocols such as ICMPv6, OSPF, UDP, TCP, Hop-by-Hop Options, and IPv6 Fragment Headers. I also learned how to use Wireshark display filters to include or exclude specific Next Header values, which made it easier to isolate and analyze different types of IPv6 traffic.

CAPSTONE LAB 2

Submitted by Hershey Beatrice Marbeda

This capstone lab covers topics learned in Chapters 5 through 8 of *Guide to TCP/IP: IPv6 and IPv4*. It assumes you are working in a Windows 7 Professional or Windows 10 Pro environment. See the Preface for details about system requirements. To complete this lab, you must have Wireshark for Windows software installed and the trace (data) file named capstone-labs-2-trace-file.pcapng.

Capstone Lab 2.1: Configure a Wireshark Display Filter to View ICMPv6 Packet Too Big messages

Time Required: 15 minutes

Objective: Use Wireshark to view a trace file and configure a display filter to view ICMPv6 Packet Too Big messages.

Description: In this lab, you will create a display filter in Wireshark and apply it to a trace file to view ICMPv6 messages. ICMPv6 Packet Too Big messages come from routers along the path between the sending host and destination host. A router in the path may have a lower maximum transmission unit (MTU) value set. Therefore, if a packet is received that is larger than the router's MTU, it sends a Packet Too Big message back to the sending host informing it of the MTU size, so the sender can start over with a smaller packet size. You need the capstone-labs-2-trace-file.pcapng file to complete the lab.

1. Start Wireshark and open the **capstone-labs-2-trace-file.pcapng** file.
2. Configure a display filter to find all ICMPv6 type 2 packets.
3. Apply the filter to the trace file.
4. Browse the list of ICMPv6 Packet Too Big packets and select one.
5. Expand **Internet Control Message Protocol v6** in the packet details pane.
6. Note the value for **MTU:** _____
7. View the **Internet Protocol Version 6** section (under the Internet Control Message Protocol v6 section) and note the source and destination IPv6 addresses. Why are they different than the source and destination addresses in the header of this packet?
8. Review additional packets from the packets list pane. Are there different source/destination pairs in this trace file?
9. What did you accomplish in this part of Capstone Lab 2? Summarize your answer.

Submitted by Hershey Beatrice Marbeda

10. Clear the display filter box.

Question 2 – Wireshark Display Filter



Question 6 – Please provide a screenshot showing MTU value

MTU	Source Address	Destination Address
1350	2001:db8:1ab:1001::109	2001:db8:1ab:3001::106

Question 7 – Please provide screenshot showing the IPv6 addresses.

MTU	Source Address	Destination Address
1350	2001:db8:1ab:1001::1	2001:db8:1ab:3001::106
1350	2001:db8:1ab:1001::108	2001:db8:1ab:3001::103
1350	2001:db8:1ab:1001::109	2001:db8:1ab:3001::103

Question 7 – Please answer the question.

The destination address of the original IPv6 packet becomes the source address of the ICMPv6 error message because the error is sent back to the original sender.

Question 8 – Please answer question.

Yes, some ICMPv6 packets have different source and destination pairs because they are related to different IPv6 communications

Question 9 – Please answer lab summary question.

Submitted by Hershey Beatrice Marbeda

In this lab, I learned how to use the proper Wireshark filter to display ICMPv6 Packet Too Big messages. I also analyzed the packets and learned why the IPv6 source and destination addresses in the ICMPv6 message can differ from the addresses in the original IPv6 header.

Capstone Lab 2.2: Create Wireshark Display Filters to View Neighbor Discovery Protocol Messages

Time Required: 15 minutes

Objective: Use Wireshark to view a trace file and configure display filters to view Neighbor Discovery Protocol messages from a specific host.

Description: In this lab, you will use Wireshark to configure specific display filters to view individual Neighbor Discovery Protocol messages. During the lab, you will discover which router is the IPv6 router (default gateway) and view validation messages that verify that the router is still available for use. You need the capstone-labs-2-trace-file.pcapng file to complete the lab.

1. In Wireshark, open the **capstone-labs-2-trace-file.pcapng** file, if it's not already open.
2. Configure a display filter so find all ICMPv6 type 133 Router Solicitation packets.
3. Apply the filter to the trace file.
4. With the Router Solicitation packets displayed, select one of those packets and note the packet number and the IPv6 source address of the host.
5. Configure a different display filter to display all IPv6 packets.
6. Apply the filter to the trace file.
7. By selecting a Router Solicitation packet and looking at all of the IPv6 messages, you can see associated packets in the vicinity of the selected Router Solicitation. You should observe a few Router Advertisements soon after the Router Solicitation message sent by the host. We specifically know that our correct IPv6 router is known by fe80::f252. Select one of the Router Advertisement packets that has an IPv6 source address of fe80::f252 and make note of this packet number.
8. Expand **Internet Control Message Protocol v6** and view the **ICMPv6 Option (Source link-layer address)** field. Note the MAC address for the router.
9. The host must know its IPv6 router's IP address and MAC address in order to communicate off its local network segment. However, the host needs to periodically verify that the IPv6 router is still

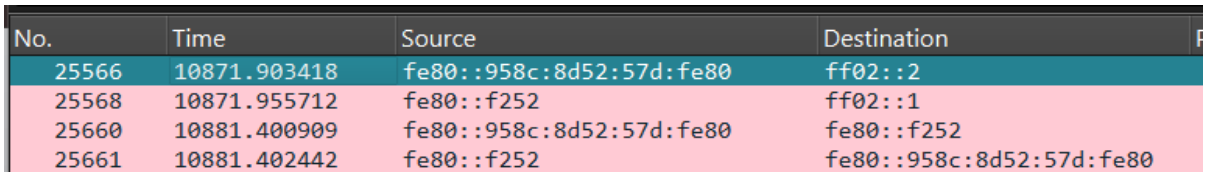
available on this network segment. This process uses a Neighbor Solicitation message from the host to the router. The router will reply back to the host with a Neighbor Advertisement message. Look through the packet list for a Neighbor Solicitation message with a source address of the host and destination address for the router, and the router's reply. Make notes as to these packet numbers. Approximately how many seconds elapsed between the Router Solicitation/Router Advertisement messages and the Neighbor Soliciation/Neighor Advertisement messages?

10. What did you accomplish in this part of Capstone Lab 2? Summarize your answer.
11. Clear the display filter box.

Question 2 – Wireshark Display



(icmpv6.type == 133 && ipv6.src == fe80::958c:8d52:57d:fe80) || (icmpv6.type == 134 && ipv6.src == fe80::f252) || (icmpv6.type == 135 && ipv6.src == fe80::958c:8d52:57d:fe80) || (icmpv6.type == 136 && ipv6.src == fe80::f252)



No.	Time	Source	Destination
25566	10871.903418	fe80::958c:8d52:57d:fe80	ff02::2
25568	10871.955712	fe80::f252	ff02::1
25660	10881.400909	fe80::958c:8d52:57d:fe80	fe80::f252
25661	10881.402442	fe80::f252	fe80::958c:8d52:57d:fe80

Question 8 – Please provide a screenshot

Router Solicitation: No. 25566, Time: 10871.903418 seconds, Host: fe80::958c:8d52:57d:fe80,

Router Advertisement: No. 25568, Time:10871.955712 seconds, Router: fe80::f252

Question 9 – Please provide screenshot of the reply.

Neighbor Solicitation: No. 25660, Time:10881.400909 seconds, Host: fe80::958c:8d52:57d:fe80

Neighbor Advertisement: No. 25661, Time:10881.402442 seconds, Router: fe80::f252

Question 9 – Please answer the question

Submitted by Hershey Beatrice Marbeda

Time elapsed between RA and RS: $10871.955712\ seconds - 10871.903418\ seconds = 0.052294\ seconds \approx 52\ ms$

Time elapsed between NA and NS: $10881.402442\ seconds - 10881.400909\ seconds = 0.001533\ seconds \approx 1.5\ ms$

Time elapsed between RA and NS: $10881.402442\ seconds - 10871.903418\ seconds = 9.499024\ s \approx 9.5\ s$

Question 10 – Please answer lab summary question.

In this lab, I learned how to display the different ICMPv6 Types: Router Advertisement, Router Solicitation, Neighbor Advertisement, and Neighbor Solicitation. Although this lab was the most confusing to me, I was able to work through the packets and display the correct information. This helped me better understand how a host discovers an IPv6 router and later verifies that the router is still reachable on the network.

Capstone Lab 2.3: Create a Wireshark Display Filters to View Multiple IPv6 Routers and DHCPv6 Servers

Time Required: 15 minutes

Objective: Use Wireshark to view a trace file and configure display filters to view Router Advertisement and DHCPv6 Advertise packets.

Description: In this lab, you will use Wireshark to configure display filters to view Router Advertisement and DHCPv6 Advertise packets. You are to investigate the configurations of various DHCPv6 messages that are being seen by the client, as well as why the client gets a DHCPv6 address from a specific DHCPv6 server, and sometimes from a different DHCPv6 server. You need the capstone-labs-2-trace-file.pcapng file to complete the lab.

1. In Wireshark, open the **capstone-labs-2-trace-file.pcapng** file, if it's not already open.
2. Configure a display filter so find ICMPv6 type 133 Router Solicitation or ICMPv6 type 134 router advertisement or DHCPv6 packets. You will need to observe these different packets and well as look at some detail in the packets.
3. Apply the filter to the trace file.

Submitted by Hershey Beatrice Marbeda

4. Early in this trace file you will observe the DHCPv6 Solicitation/Advertise/Request/Reply (SARR) process, but there are two DHCPv6 Advertise messages seen. The DHCPv6 Advertise message from fe80::f252 is considered the correct one for this network. The issue is, which DHCPv6 server will the host request an address from? The answer is, whichever DHCPv6 has the higher priority setting in its Advertise message, that is, the DHCPv6 server the host will request an IPv6 address from. If the DHCPv6 servers have the same value set (generally DHCPv6 servers have no preference value set and therefore none is sent in the advertise message), the host will request an address from the first DHCPv6 server that replies. Select the DHCPv6 Request message that the host sends soon after receiving the two DHCPv6 Advertise messages.
5. Expand **DHCPv6**, expand **Server Identifier**, and then view and make note of the **Link-layer address**.
6. Select the DHCPv6 Reply message that is after the DHCPv6 Request. Looking in the Ethernet II field on the packet details pane, this should be the same MAC address as noted in the previous step (link-layer address). Is this the known/valid DHCPv6 server?
7. In the packet list, scroll down to and select packet number **9679**, which is a DHCPv6 Advertise message.
8. Expand **DHCPv6**. Looking at those options, do you see a Preference field?
9. Select the next packet (9680) and view the **DHCPv6** options. Do you see a Preference field?
10. Select the DHCPv6 Request message that the host sends soon after receiving the two DHCPv6 Advertise messages.
11. Expand **DHCPv6**, expand **Server Identifier**, and then view and make note of the **Link-layer address**.
12. Select the DHCPv6 Reply message that is after the DHCPv6 Request. Looking in the **Server Identifier** field in the packet details pane, this should be the same MAC address as noted in the previous step (link-layer address). Is this the known/valid DHCPv6 server (or its relay)?
13. Bonus question: Is fe80::f252 the actual DHPCv6 server? How can you tell?
14. What did you accomplish in this part of Capstone Lab 2? Summarize your answer.
15. Clear the display filter box.

Question 2 – Wireshark Filter Display



Submitted by Hershey Beatrice Marbeda

Question 4 – Please provide screenshot

DHCPv6 Request: Link-layer address: 00:0c:29:02:5f:8c

DHCPv6 Reply: Link-layer address: 00:0c:29:02:5f:8c

Question 6 – Please answer question.

Yes, the MAC addresses match, but it is not the known valid DHCPv6 server because its IPv6 address is not fe80::f252

Question 8 – Please answer question.

No

Question 9 – Please answer question.

Yes

Question 12 – Please answer question.

Link-layer address: 000c29c10d71, Source Address: fe80::f252

Yes, because the DHCPv6 Reply has a source address of fe80::f252, which is the known valid DHCPv6 server or relay

Bonus question 13 (Optional) – please answer the question

Yes, fe80::f252 is the actual DHCPv6 server because it is the device that sent the DHCPv6 Advertise offer message to the client, which is fe80::958c:8d52:57d:fe80.

Question 14 (required) – Please answer lab summary question.

In this lab, I learned how to analyze and confirm ICMPv6 types 133 and 134, as well as DHCPv6 Advertisement and Reply messages. I also learned how to verify the correct router and DHCPv6 server by checking IPv6 Addresses, MAC Addresses, Server Identification information, and DHCPv6 preference values.

Submitted by Hershey Beatrice Marbeda

Capstone Lab 2.4: Configure Wireshark Display Filters to View DNS Queries for AAAA Record Types

Time Required: 15 minutes

Objective: Use Wireshark to view a trace file and configure display filters to view DNS queries for AAAA Record types specifically from IPv6 hosts.

Description: In this lab, you will create display filters in Wireshark and apply them to a trace file to view DNS messages. You need the capstone-labs-2-trace-file.pcapng file to complete the lab.

1. In Wireshark, open the **capstone-labs-2-trace-file.pcapng** file, if it's not already open.
2. Configure a display filter to find all DNS packets.
3. Apply the filter to the trace file.
4. Browse the list of DNS packets and select a packet that has a "AAAA" query in the Info column.
5. Expand **Domain Name System (query)**.
6. Expand **Queries**, expand the listed DNS section, right-click **Type: AAAA (IPv6 Address) (28)**, select **Apply as Filter**, and then select **Selected**.
7. In the packet list pane, select a query packet with a source IPv6 address of 2001:db8:1ab:1001::108, expand **Internet Protocol Version 6**, right-click on **Source: 2001:db8:1ab:1001::108**, select **Apply as Filter**, and then select **…and Selected**.
8. In the Domain Name System (query) section, expand **Queries** (if not already expanded), expand the listed domain name section, right-click **Name: <domain-name>**, select **Apply as Filter**, and then select **…and Selected.**
9. In the filter bar, edit the current domain-name to read **www.ipv6sandbox.com**, and press **Enter**.
10. How many packets are displayed?
11. What did you accomplish in this part of Capstone Lab 2? Summarize your answer.
12. Close Wireshark, and do not select **Save** if prompted to save changes upon closing.

Question 2 – Wireshark Filter Display



Question 10 – Please answer the question.

Submitted by Hershey Beatrice Marbeda

66 Packets



Question 11 – Please answer the lab summary question.

In this lab, I learned how to filter out a single IPv6 address and specified a domain using dns.qry.type==28 (for IPv6, AAAA) and dns.qry.name==”ipv6sandbox.com”. Overall, I used logic gates and my knowledge to filter out the information I needed to investigate.

Submitted by Hershey Beatrice Marbeda